



Wing geometry and kinematic parameters optimization of bat-like robot fixed-altitude flight for minimum energy

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ARTICLE INFO

Article history:

Received 4 October 2022

Received in revised form 26 March 2023

Accepted 22 June 2023

Available online 28 June 2023

Communicated by Huihe Qiu

Keywords:

Bat-like robot

Fixed-altitude flight

Minimum energy

Combined optimization

ABSTRACT

In this paper, the geometry parameters (WGP) and kinematic parameters (WKP) of bat wings are optimized by using the revised quasi-steady aerodynamic model and genetic particle swarm optimization algorithm (GPSO) according to the structural characteristics and kinematic law of bat, to meet the requirement of minimum flapping energy and improve the endurance of bat-like robot. For the first time, the dynamic model of bat flapping motion is improved by considering the effects of active deformation, passive flexible deformation, gravity moment, and inertia moment of the wing structure. Based on the power density model, the optimization objective function is established, and the lift-weight ratio (LtoW), Reynolds number (Re), aspect ratio (AR), Strouhal number (St), and boundary constraints of each parameter are added to complete the optimization of WGP and WKP. The optimized aerodynamic forces and power densities are compared with the estimates for actual bats flying at a fixed-altitude, and the results showed that the optimized parameters resulted in more stable aerodynamic forces and reduced energy consumption by half. Finally, the sensitivity analysis of the influence of each parameter on the LtoW and power density is carried out to better understand the effect of each parameter on keeping fixed-altitude flight and reducing energy consumption. This study can provide the theoretical basis for reasonable parameter design of bat-like robot.

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1. Introduction

Since the 1990s, with the development of aerodynamics theory and new materials, flying robots have entered the trend of development. All kinds of flying robots have been studied all over the world [1]. Flying robot flight modes include fixed wing, rotor wing, flapping wing, etc. Among them, due to the bionic shape conditions, the flapping wing robot has good invisibility, low noise, and high maneuverability, and it has great prospects in detection and cruising [2]. The biggest demand of human beings for bionic flapping wing robots is to improve the concealability of an investigation by using its ultra-high bionic shape conditions and to realize the long-term fixed-altitude flight in a specific area to complete the investigation and cruise missions. This requires a bionic flapping wing flying robot to meet the vivid shape conditions, as well as the minimum energy consumption requirements of fixed-altitude flight [3]. However, in the flapping wing robot design process, completely according to the WGP and WKP of the actual

biological design of a bionic flying robot [4], the aerodynamic characteristics cannot support the robot to carry electronic equipment to meet specific functions and a certain quality of the battery to ensure flight distance and endurance [5]. Therefore, how to efficiently imitate the physiological structure and kinematic laws, especially those with excellent flight performance and maintaining fixed-altitude flight, and how to rationally optimize the WGP and WKP of the bionic flapping wing robot, so that it has the minimum energy consumption while maintaining fixed-altitude flight, to improve its endurance is the current research hotspot [6].

Among the many flying creatures in nature, bats are known as 'flying masters'. They can not only keep fixed-altitude flight but also have better mobility than birds of the same size [7]. To a great extent, it meets the functional requirements of the bionic flapping wing robot, and once became one of the main imitation objects of flapping wing robot design [8]. Related research on the bat-like robot has attracted widespread attention from researchers around the world [9]. Representative achievements are mainly: Memory alloy joint bat-like robot of Pontificia Universidad Javeriana [10], Brown University's multi degrees of freedom bat-like robot [11,12], B2 developed by the University of Illinois and California Institute of Technology [13,14], Bionic Flying Fox of Festo, etc. However, the

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Nomenclature

AR	Aspect ratio	P^*	Power density
c	Chord length	r	Distance of strip from flapping axis
C_D	Drag coefficient	Re	Reynolds number
C_L	Lift coefficient	St	Strouhal number
C_r	Rotational circulation coefficient	v	Relative velocity of the strip
C_{rs}	Rotational damping coefficient	$v_{f\infty}$	Freestream flow velocity
El_{lead}	Bending rigidity of the leading edge	α	Relative angle of attack
El_{back}	Bending rigidity of the trailing edge	ω_θ	Flapping angular velocity
I	Wingtip shape index	ω_t	Passive torsional angular velocity
$LtoW$	Lift weight ratio	θ_0	Amplitude of flapping Angle
$O_b-X_bY_bZ_b$	Body coordinate system	φ	Passive torsion angle
$O_w-X_wY_wZ_w$	Wing coordinate system	ν_{air}	Kinematic viscosity coefficient of air
$O_m-X_mY_mZ_m$	Membrane coordinate system	ρ	Density of the air
P_{total_aero}	Pneumatic power of flapping	β	Angle between the flapping plane and the horizontal plane
P_{total_stru}	Power of structure		

existing bat-like robot mentioned above only studies the physiological structure and kinematic law of bats, but does not consider the problem of the minimum energy consumption when flying at a fixed-altitude. Therefore, the bat-like robot studied above has good bionic performance and kinematic performance, but may not be suitable for completing missions such as investigation and cruising. In recent years, many scholars have also studied the minimum energy consumption of the flapping wing robot when flying at a fixed-altitude (hover) [15], but all of the research is aimed at micro flapping wing robots. Ke [16] used the quasi-steady aerodynamic model and the traditional hybrid genetic algorithm to optimize the WGP and WKP of the fruit fly so that it can obtain the minimum energy consumption to maintain hover flight. Yan [17] optimized the WKP of the micro flapping wing robot by using the four dynamic models of leading edge vortex, rotational lift, non-circulatory contributions and flow unsteadiness to obtain the parameter model with the minimum energy consumption during hovering. Nabawy [18] proposed to reduce energy consumption by fixing the WGP and the flapping frequency, only changing the flapping angle. By calculating, it is concluded that the flapping angles of sinusoidal and rectangular shapes are beneficial to minimize energy. Shan [19] established an electric-piezo-aeroelastic coupling lumped-parameter model, which can predict the dynamic behavior of strain-actuated flapping wings, and drive wing flapping through a piezoelectric actuator which can reduce the weight of the model and reduce energy consumption compared to conventional systems [20]. Similarly, Mukherjee uses electric-piezo devices to drive the movement of dragonfly wings [21]. However, the above research results of maintaining fixed-altitude flight for minimum energy are all based on the model of micro flapping wing robots. The micro flapping wing robots are small in size and light in weight and can ignore the influence of structural and mechanical characteristics of wings, so only the aerodynamic characteristics of wings need to be considered. Moreover, micro flapping wing robots generally use materials with a large elastic modulus (3–4 GPa) such as polyimide (PET) [22] as wings, ignoring the influence of flexible deformation on aerodynamic characteristics and energy consumption. For the large bat-like flapping robot, because of its large wing size and complex deformation structure, the mass force and inertia force of the wing cannot be ignored. In addition, bat-like flapping wing robots generally use materials with a small elastic modulus (1–3 MPa) similar to PDMS [13] as wings, so the wings have large flexible deformation, and the influence of this deformation should be considered in dynamic modeling. Given the above problems, this paper establishes a parameter calculation model of deformable flexible wings based on the structural characteristics

and kinematic laws of actual bats and considers the flexible deformation of the wing membrane and the influence of the gravity and inertia force of the wings. Based on the improved quasi-steady theory, a dynamic model of wing flapping suitable for bat-like robots is established. Based on the dynamic model, the power density model is established, and a new GPSO is used to optimize the WGP and WKP with the minimum energy consumption under the condition of fixed-altitude flight, combined with $LtoW$, Re , AR , St , and boundary constraints of the wing. In addition, the sensitivity analysis of the influence of each parameter on the $LtoW$ ratio and power density is performed to better understand the effect of each parameter on maintaining fixed-altitude flight and reducing energy consumption.

The rest of the paper is organized as follows. Firstly, in Section 2, the structure configuration and wing movement law of bat-like robot are introduced. Then, in Section 3, the aerodynamic model and mechanics model of the flapping movement of the wing are established. In Section 4, the power density model is given, and the optimization model for WGP and WKP is established by combining GPSO. Then, in Section 5, the optimization results of WGP and WKP with the minimum energy consumption are presented, and the power contribution of the dynamic model and the sensitivity analysis of each WGP and WKP are analyzed in detail. Finally, the conclusion is drawn in Section 6.

2. Analysis of the configuration and motion law of the wing

According to the characteristics of the bat, this paper carries out the bionic design of the structure and the modeling of the movement mode, so that it can realize the movement form of wings unfolding when flapping downstroke and folding when flapping upstroke. The deformation of the flexible wing studied in this paper includes two parts: active deformation and passive deformation. Among them, the passive deformation is realized by the flexible deformation of the membrane, the active deformation is realized by the folding and unfolding of the watt six-bar linkage mechanism to drive the wing to move, as shown in Fig. 1. In the figure, three colors of yellow, red and blue are used to show the movement of the wing and the deformation of the membrane during the initial position of the wing, the downstroke and the upstroke stage.

To carry out the aerodynamic calculation, the coordinate system of the bat-like robot is established, as shown in Fig. 2 The Body coordinate system $O_b-X_bY_bZ_b$ is fixed at the center of gravity of the Body. The coordinate system of the flapping wing $O_w-X_wY_wZ_w$ is located at the leading edge shoulder of the wing. The coordinate system $O_m-X_mY_mZ_m$ is located on the membrane. Since the

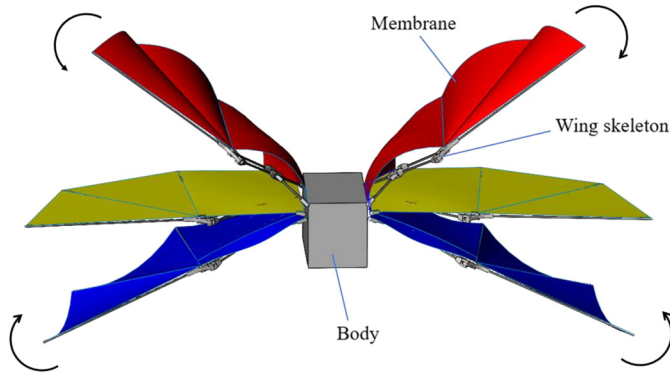


Fig. 1. The movement and deformation of wing. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

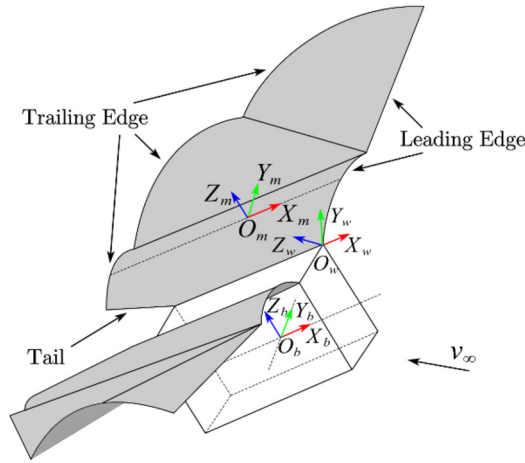


Fig. 2. Coordinate system definition.

membrane is made of thin film, it belongs to the symmetric membrane, and the pressure center of the symmetric membrane is also the aerodynamic center [23], so the origin O_m is located at the pressure center.

The watt six-link structure diagram with wings folding and unfolding is shown in Fig. 3. In the figure, point P_0 is located at the origin O_w of the wing flapping coordinate system ($O_w-X_wY_wZ_w$). Points $P_1 \sim P_5$ are the connection points between the wing skeleton and the membrane, and point P_7 is the movement point that drives the deformation of the wing skeleton. The variable y_{p2} marked in the figure represents the displacement of wing deformation movement; The red and green arrows indicate the direction of movement of the wing skeleton.

3. Model of aerodynamic forces and moments

The membrane will deform with the movement of the wing, so the chord length of each position is different at each time. It is necessary to use BEM to divide the strip of the membrane and calculate the chord length of each strip. The membrane is evenly divided into 1000 strips along the direction from the flapping axis to the wing tip. Fig. 4 shows the structure of the membrane and the schematic diagram of the membrane plane. The shape of the membrane is shown on the left side of the flapping axis, and the strip division is shown on the right side of the flapping axis.

3.1. Related geometric parameters of membrane

In the stage of bat flapping, the membrane will produce a certain degree of spatial deformation due to its movement and elas-

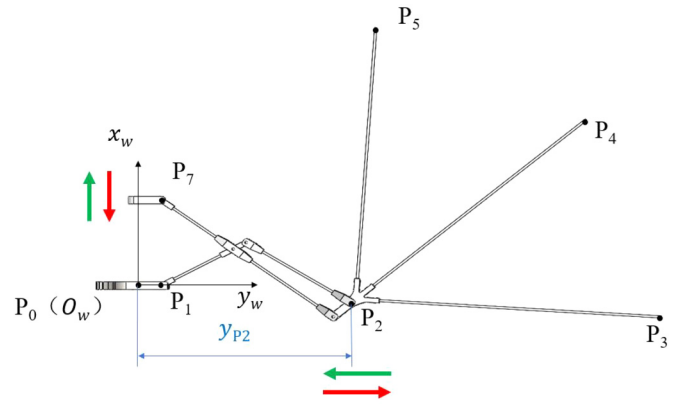


Fig. 3. Wing deformation mechanism diagram.

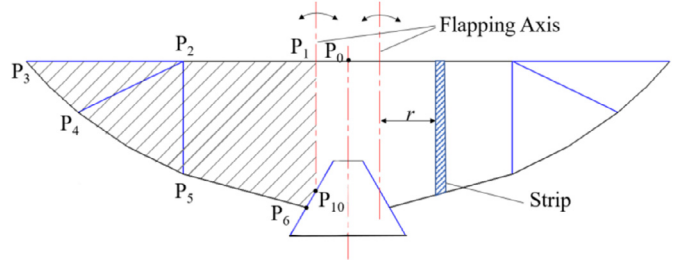
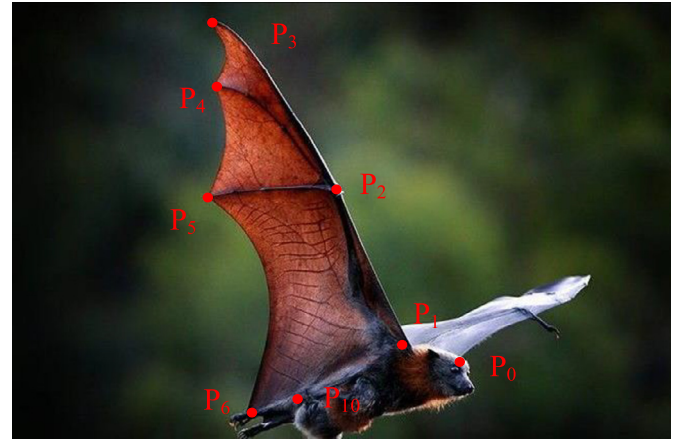


Fig. 4. Shape and strip division of membrane.

ticity, which can be divided into two parts: The first part is that because the wing skeleton flaps around the flapping axis, but the tail remains stationary, there will be an angle between the two, but the membrane is stuck to the wing skeleton and tail at the same time, so there will be a height difference between the leading edge and the trailing edge of the membrane relative to the initial plane. The second part is the elastic deformation of the membrane because of the aerodynamic force. To facilitate understanding, the deformation of the membrane is decomposed for illustration, as shown in Fig. 5.

Fig. 5 (a) represents the three-dimensional model diagram of the membrane in the downstroke, including the deformation of the membrane and the projection of the membrane on the initial plane (projection is represented by dotted lines). Fig. 5 (b) shows the model in the downstroke of the membrane, which is used to explain the deformation of the membrane, including the deformation diagram of the membrane and the projection diagram of the initial plane; Fig. 5 (c) and Fig. 5 (d) are schematic diagrams of the deformation of the membrane in Fig. 5 (b) after decomposition (separated along the plane $P_1P_2P_3P_4P_5P_6P_{10}$), where Fig. 5 (c)

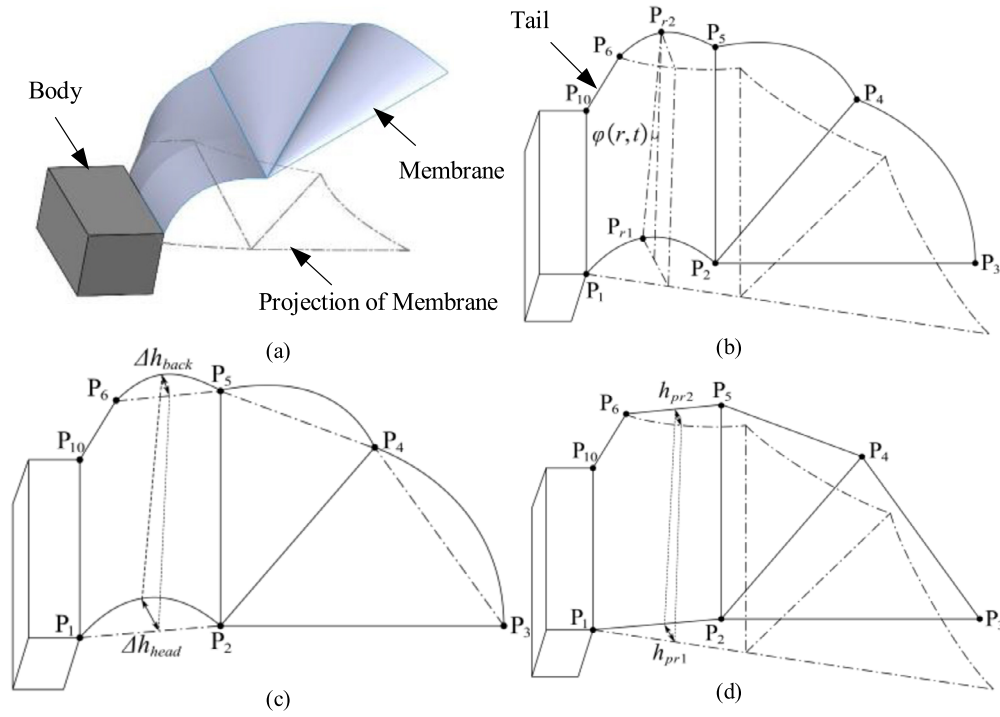


Fig. 5. The decomposition of membrane deformation.

represents the elastic deformation of the membrane because of the aerodynamic force; Fig. 5 (d) represents the height difference between the torsional deformation of the leading and trailing edges of the membrane relative to the initial plane (the plane composed of dotted lines).

In Fig. 5, P_6 and P_{10} are fixed on the bat's tail and do not change position with the flapping of wings (the tail and body are in the same plane). $\varphi(r, t)$ is the angle of the strip (the strip with a distance r from the flapping axis P_1P_{10} at time t) relative to the initial plane, which is defined as the passive torsion angle. Its physical significance is the angle between the strip string and the flapping axis after the membrane twisted around the leading edge axis. h_{pr1} is the height of the leading edge due to flapping; h_{pr2} is the height of the trailing edge due to flapping; Δh_{head} is the height of the leading edge due to the flexible deformation of the membrane; Δh_{back} is the height of the trailing edge due to the flexible deformation of the membrane. The above deformation parameters ($\varphi(r, t)$, h_{pr1} , h_{pr2} , Δh_{head} , Δh_{back} , etc.) all represent the deformation of the membrane, so these variables are located on the membrane plane and in the membrane coordinate system ($O_m-x_m y_m z_m$).

Due to the particularity of the bat wing structure, the deformation of the side membrane fixed to the tail, the arm membrane of the middle section, and the hand membrane of the wing hand part are different, so the research needs to be divided into three parts [24], as shown in Fig. 6.

The first part ($P_1P_{10}P_6P_{11}$) is the part connected with the membrane and the body, in which P_1P_{10} is connected with the body and $P_{10}P_6$ is connected with the tail, both of which do not produce deformation, so this part only needs to calculate the deformation of the leading edge; The second part ($P_{11}P_6P_5P_2$) is the middle part of the wing, in this part, two equivalent deformations should be considered and calculated simultaneously; The third part ($P_2P_5P_4P_3$) is the part where the membrane is connected with the wing hand, this part only needs to calculate the deformation of the trailing edge.

First, for the first part, the analytical diagram of bat membrane deformation is shown in Fig. 7. In the figure, r represents the dis-

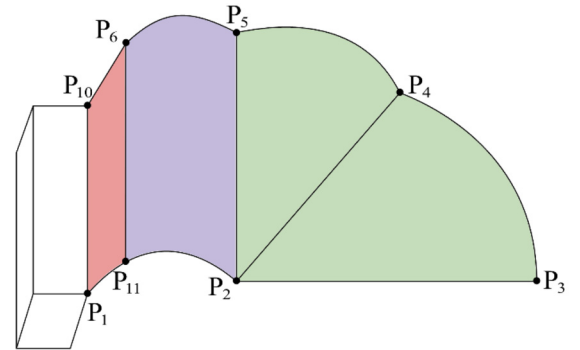


Fig. 6. The membrane is divided into three parts.

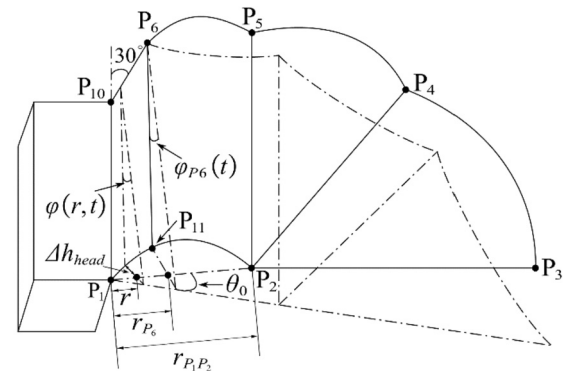


Fig. 7. Analytical diagram of bat membrane deformation.

tance between the strip and the flapping axis P_1P_{10} ; $\varphi(r, t)$ is the passive torsion angle of the strip at this moment where the distance from the flapping axis is r ; r_{P6} represents the distance between the strip P_6P_{11} and the flapping axis; $\varphi_{P6}(t)$ represents the passive torsion angle of the strip where P_6P_{11} is located, which is also the largest passive torsion angle (absolute value) of the first

part; θ_0 is the flapping angle; Δh_{lead} is the height formed by the flexible deformation of the membrane from the leading edge of the strip at a distance r from the flapping axis.

P_1P_2 is the leading edge of the membrane, which will be deformed by aerodynamic force. For calculation, P_1P_2 is assumed to be a beam structure (a beam with a large amount of elastic deformation) with both ends fixed instantaneously. The beam is subjected to a uniform force (assuming the aerodynamic force as uniform force), which will produce bending deformation, and its height due to elastic deformation is:

$$\Delta h_{lead} = \frac{\frac{q}{2} \left(\frac{1}{12} r^4 - \frac{r_{P1P2}}{6} r^3 \right) + \frac{q r_{P1P2}^3}{24}}{EI_{lead}} \quad (1)$$

Where EI_{lead} is the bending rigidity of the leading edge; q is the uniform aerodynamic load (the ratio of aerodynamic force to the area of membrane); r_{P1P2} is the distance between P_1 and P_2 .

Through the geometric relationship in Fig. 7, the expressions of chord length and passive torsion angle can be obtained:

$$c(r, t) = \sqrt{\left(2r \cdot \sin \frac{\theta_0}{2} - \text{sign} \left(\theta_0 \frac{d\theta_0}{dt} \right) \Delta h_{lead} \right)^2 + (P_1P_{10} + r \tan 30^\circ)^2}, \quad 0 \leq r \leq r_{P6}(t) \quad (2)$$

$$\varphi(r, t) = \pm \arctan \frac{2r \cdot \sin \frac{\theta_0}{2} - \text{sign} \left(\theta_0 \frac{d\theta_0}{dt} \right) \Delta h_{lead}}{P_1P_{10} + r \tan 30^\circ}, \quad 0 \leq r \leq r_{P6}(t) \quad (3)$$

Next, the calculation of second part ($P_{11}P_6P_5P_2$ in Fig. 6) is based on the decomposition of membrane deformation shown in (c) and (d) in Fig. 5. Similar to the deformation of the leading edge, the trailing edge part P_5P_6 is also assumed to be a beam structure with both ends fixed instantaneously, then the height of the leading and trailing edges due to elastic deformation are:

$$\Delta h_{lead} = \frac{\frac{q}{2} \left(\frac{1}{12} r^4 - \frac{r_{P1P2}}{6} r^3 \right) + \frac{q r_{P1P2}^3}{24}}{EI_{lead}} \quad (4)$$

$$\Delta h_{back} = \frac{\frac{q}{2} \left(\frac{1}{12} (r - r_{P6})^4 - \frac{r_{P5P6}}{6} (r - r_{P6})^3 \right) + \frac{q r_{P5P6}^3}{24} (r - r_{P6})}{EI_{back}} \quad (5)$$

Where EI_{back} is the bending rigidity of the trailing edge; r_{P5P6} is the distance between P_5 and P_6 ; r_{P6} is the distance between P_6 and flapping axis.

The height of any point on the leading edge and the height of the corresponding trailing edge point:

$$h_{Pr1} = \frac{h_{P2}}{r_{P1P2}} r_{Pr1} = 2r_{Pr1} \cdot \sin \frac{\theta_0}{2} \quad (6)$$

$$h_{Pr2} = \frac{r_{Pr2} - r_{P6}}{r_{P1P2} - r_{P6}} h_{P2} \quad (7)$$

The projection on horizontal plane of the chord length formed by the connecting point P_{r1} and P_{r2} :

$$P'_{r1}P'_{r2} = P_2P_5 + \left(\frac{r_{P1P2} - r_{Pr1}}{r_{P1P2} - r_{P6}} \right) \times [(r_{P6} \cdot \tan 30^\circ + P_1P_{10}) - P_2P_5] \quad (8)$$

The height difference between P_{r1} and P_{r2} is:

$$h_{Pr2Pr1} = \left(h_{Pr2} - \text{sign} \left(\theta_0 \frac{d\theta_0}{dt} \right) \Delta h_{back} \right) - \left(h_{Pr1} - \text{sign} \left(\theta_0 \frac{d\theta_0}{dt} \right) \Delta h_{lead} \right) \quad (9)$$

So, the expressions of chord length and passive torsion angle can be obtained:

$$\varphi(r, t) = \text{atan} \left(\frac{h_{Pr2Pr1}}{P'_{r1}P'_{r2}} \right), \quad r_{P6} \leq r \leq r_{P1P2} \quad (10)$$

$$c(r, t) = P_{r1}P_{r2} = \sqrt{h_{Pr2Pr1}^2 + (P'_{r1}P'_{r2})^2}, \quad r_{P6} \leq r \leq r_{P1P2} \quad (11)$$

Finally, the third part ($P_2P_5P_4P_3$ in Fig. 6) is connected with the wing hand, and the skeleton layout of the wing hand is fixed, so the chord length of this part is related to the shape of the wingtip.

For this part, the height of the trailing edge due to elastic deformation is:

$$\Delta h_{hand} = \frac{\frac{q}{2} \left(\frac{1}{12} (\Delta r)^4 - \frac{1}{6} \frac{r_{P2P3}}{\cos \varphi_{hand}} (\Delta r)^3 \right) + \frac{q \cdot \Delta r}{24} \left(\frac{r_{P2P3}}{\cos \varphi_{hand}} \right)^3}{EI_{hand}} \quad (12)$$

Where:

$$\Delta r = \frac{r - r_{P1P2}}{\cos \varphi_{hand}} \quad (13)$$

$$\cos \varphi_{hand} = \frac{r_{P2P3}}{\sqrt{(r_{P2P3})^2 + (r_{P2P5})^2}} \quad (14)$$

So, the expressions of chord length and passive torsion angle can be obtained:

$$c(r, t) = \sqrt{\left(P_2P_5 \cdot \left(1 - \left(\frac{r - r_{P1P2}}{r_{max} - r_{P1P2}} \right)^I \right) \right)^2 + (\Delta h_{hand})^2}, \quad r \geq r_{P1P2} \quad (15)$$

Where I is the wingtip shape index (I reflects the wingtip shape).

$$\varphi(r, t) = \text{atan} \left(\frac{\Delta h_{hand}}{P_2P_5 \cdot \left(1 - \left(\frac{r - r_{P1P2}}{r_{P2P3}} \right)^I \right)} \right), \quad r \geq r_{P1P2} \quad (16)$$

In summary, the passive torsion angle of the wing membrane is:

$$\varphi(r, t) = \begin{cases} \pm \arctan \frac{2r \cdot \sin \frac{\theta_0}{2} - \text{sign} \left(\theta_0 \frac{d\theta_0}{dt} \right) \Delta h_{lead}}{P_1P_{10} + r \tan 30^\circ}, & 0 \leq r \leq r_{P6}(t) \\ \text{atan} \left(\frac{h_{Pr2Pr1}}{P'_{r1}P'_{r2}} \right), & r_{P6} \leq r \leq r_{P1P2} \\ \text{atan} \left(\frac{\Delta h_{hand}}{r_{P2P5} \cdot \left(1 - \left(\frac{r - r_{P1P2}}{r_{P2P3}} \right)^I \right)} \right), & r \geq r_{P1P2} \end{cases} \quad (17)$$

The chord length distribution of the wing surface $c(r, t)$ can be summarized as:

$c(r, t)$

$$= \begin{cases} \sqrt{(2r \cdot \sin \frac{\theta_0}{2} - \text{sign}(\theta_0 \frac{d\theta_0}{dt}) \Delta h_{lead})^2 + (P_1 P_{10} + r \tan 30^\circ)^2}, & 0 \leq r \leq r_{P6}(t) \\ \sqrt{h_{Pr2Pr1}^2 + (P'_{r1} P'_{r2})^2}, & r_{P6} \leq r \leq r_{P1P2} \\ \sqrt{(r_{P2P5} \cdot (1 - (\frac{r-r_{P1P2}}{r_{max}-r_{P1P2}})^I))^2 + (\Delta h_{hand})^2}, & r \geq r_{P1P2} \end{cases} \quad (18)$$

3.2. Aerodynamic modeling of wing membrane

3.2.1. The composition of the aerodynamic model

In this paper, assuming that the bat-like robot always flies at a constant speed and stably, the quasi-steady aerodynamics model can be used to predict the aerodynamics of the flapping membrane. What's more, the blade element method (BEM) is used to strip the membrane, the unsteady flow is regarded as quasi-steady flow at each time and in each strip. Whitney and Wood [25] have proved that the BEM can ensure the applicability of the quasi-steady aerodynamic model. These aerodynamic and aerodynamic moment are derived from translational circulation, rotational circulation, aerodynamic damping, and added-mass force.

The translational circulation generated by the membrane consists of two parts, one is the circulation generated by the free-stream flow acting on the membrane. The other part is the change of the velocity caused by the flapping of the wing, resulting in the circulation. Using the data of the bat lift coefficient and drag coefficient obtained by Colorado [7], the empirical equation summarized by Dickinson [26] is modified. The revised equations are as follows:

$$C_L = -8.487 - 9.588 \sin(0.0233\alpha + 4.304) \quad (19)$$

$$C_D = 0.2844 + 0.2262 \cos(0.09426\alpha + 3.652) \quad (20)$$

The lift and drag on each strip are:

$$dL_{trans} = \frac{1}{2} \rho v^2 C_L c dr \quad (21)$$

$$dD_{trans} = \frac{1}{2} \rho v^2 C_D c dr \quad (22)$$

Aerodynamic force and aerodynamic moment about the flapping axis on each strip are:

$$dN_{trans} = dL_{trans} \cdot \cos \alpha + dD_{trans} \cdot \sin \alpha \quad (23)$$

$$dM_{trans,x} = dN_{trans} \cdot r \quad (24)$$

While the membrane is large and the material is soft, the rotational circulation cannot be ignored [27]. Therefore, the aerodynamic force caused by the passive rotation needs to be considered. The aerodynamic force and aerodynamic moment of passive rotation on any strip are:

$$dF_{rot} = \frac{1}{2} \rho \omega_\theta \omega_t C_r r c^2 dr \quad (25)$$

$$dM_{rot,x} = dF_{rot} \cdot r \quad (26)$$

Where ω_θ is the flapping angular velocity, ω_t is the passive rotation angular velocity, C_r is the rotational circulation aerodynamic force coefficient, $C_r = 0.75\pi$.

Wood [25] has proved the existence of the dissipative damping moment through experiments. It can adopt the theoretical basis of Wood to establish the damping moment model:

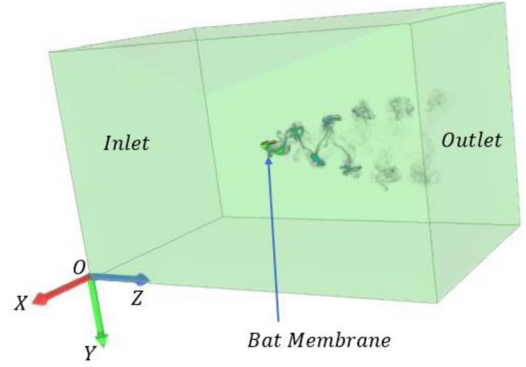


Fig. 8. Simulated wind tunnel.

$$dM_{damp,x} = -\frac{1}{2} \rho \omega_\theta^2 C_{rs} \int_0^r r^3 dr \quad (27)$$

Where C_{rs} is rotational damping coefficient, $C_{rs} = 2$.

In aerodynamic research, the non-circulation mechanism generally refers to added-mass force. The method mentioned by Wood [25] is used to derive and calculate the normal added-mass force on the plane of the membrane:

$$dF_{add,N} = \frac{1}{4} \pi \rho c^2 \ddot{\theta} r^2 dr - \frac{1}{4} \pi \rho c^2 c_{dis} \ddot{\phi} r dr \quad (28)$$

$$dM_{add,x} = \frac{1}{4} \pi \rho c^2 \ddot{\theta} r^3 dr - \frac{1}{4} \pi \rho c^2 c_{dis} \ddot{\phi} r^2 dr \quad (29)$$

Where c_{dis} is the distance between the pitch axis and the midpoint of the strip, $c_{dis} = c/2$.

Since the mass of the wing membrane is small, the influence of gravity and inertia forces on the aerodynamic force of the wing membrane can be ignored. Based on the above aerodynamic forces, the resultant aerodynamic forces can be obtained as:

$$dF_{res} = dN_{trans} + dF_{rot} + dF_{add,N} \quad (30)$$

$$dM_{total,aero,x} = dM_{trans,x} + dM_{rot,x} + dM_{damp,x} + dM_{add,x} \quad (31)$$

3.2.2. Verification of aerodynamic models

The commercial CFD software XFlow based on the Lattice Boltzmann method is used to simulate the flow around the flapping wing. Shao [28] has proved that XFlow can ensure the accuracy of flow field simulation. Without considering the coupling of the flow field, the simulation with a single wing can shorten the calculation time and improve the calculation efficiency.

Fig. 8 shows the simulated wind tunnel used in this paper. The membrane material is PDMS, this material density is 1000 kg/m³, the elastic modulus is 2.3 MPa, and the Poisson's ratio is 0.4. Abaqus is used to mesh the model. The hexahedral mesh is used for the mesh, and C3D8R is selected for the mesh element. To reduce the influence of boundary conditions on the flow field, the membrane model is placed in a wind tunnel of about 10l × 10l × 15l, and l is the length of the bat wingspan (0.3404 m). In the flow field area, Z-axis direction is set as freestream flow velocity, X-axis and Y-axis have no flow velocity, and the outlet boundary is also the freestream outflow with a velocity gradient of 0. To ensure the convergence of the results, set the Courant number to 0.1. The particle size in the wind tunnel is set as 0.01 and the total particle number is about 64000000, which is enough to ensure the accuracy of the calculation. The flow field medium chooses the air under normal temperature, the air density is 1.225 kg/m³ and the air viscosity is 1.7894e-5 Pa.s. Other simulation conditions are set

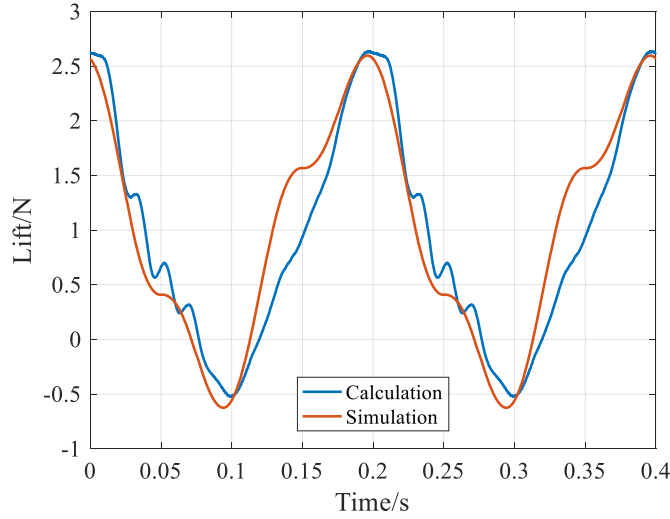


Fig. 9. The lift between the calculation and simulation.

as the maximum flapping angle of the membrane is 27.5° , the flapping frequency is 5 Hz, the angle of attack of the membrane is 15° , and the freestream flow velocity is 5 m/s.

The lift measured by simulation is compared with the calculated result shown in Fig. 9.

The correctness of the theory can be verified through the analysis of Fig. 9. The trends of the two are generally the same, and the curve fitting is very high. From the numerical results, the maximum lift calculated by theory is 2.596 N, and the maximum lift obtained by simulation is 2.637 N, the error between the two is 2.59%, within an acceptable range; The average lift calculated by theory is 1.000 N, and the average lift obtained by simulation is 0.9502 N, the error between the two is 4.974%, within an acceptable range. From what has been discussed above, the theoretical model can be considered accurate.

3.3. Mechanics modeling of wing structure

Since the materials selected for each part are different (for example, aluminum alloy is used for the shoulder, nylon is used for the joint, and carbon fiber is used for the connecting rod), the calculation needs to be carried out separately, and the wing structure in Fig. 3 is detailed and explained, as shown in Fig. 10.

Because the length of each joint component is short compared with the wingspan, and each joint component is almost symmet-

rical, the joint parts such as shoulder, elbow, and wrist can be regarded as particles, and the positions are all located at the given nodes. The center of gravity of the carbon fiber tube is located at the center of the tube (the midpoint of two adjacent nodes), such as the carbon fiber tube of Arm1, the center of gravity is:

$$(x_{10}, y_{10}) = \left(\frac{x_{P1} + x_{P9}}{2}, \frac{y_{P1} + y_{P9}}{2} \right) \quad (32)$$

The gravity moment of the wing structure (17 parts in total) can then be calculated using the following equation:

$$dM_{w2f-x} = - \sum_{i=0}^{16} m_i \cdot g \cdot x_i \sin \beta \cos \theta_0 \quad (33)$$

Where β is the angle between the flapping plane and the horizontal plane.

Some scholars have conducted experimental studies on aerodynamic forces and found that inertia force plays an important role in aerodynamic test experiments, especially in the vicinity of wing pitch inversion, where the inertial force is almost equivalent to the additional mass force [16]. Therefore, it is necessary to consider the contribution of inertia force.

Similar to the assumption of gravity moment, when calculating the inertia moment of the wing skeleton structure, the moment of inertia changes little when rotation occurs because the joint components are short, so the moment of inertia is regarded as a constant. The carbon fiber tube has a long length, and its moment of inertia changes greatly when it rotates, which cannot be ignored. Therefore, the finite element idea is used to divide the carbon fiber tube into n parts and calculate the inertia moment of each part. The inertia moment of the wing structure (10 joint and 7 carbon fiber tube) can then be calculated using the following equation:

$$\begin{aligned} dM_{i2f_x} &= - \left(\sum_{i=0}^9 (J_i + m_i \cdot x_i^2) + \sum_{i=10}^{16} \sum_{j=1}^n m_{ij} \cdot x_{ij}^2 \right) \cdot \alpha_0 \\ &= - \left(\sum_{i=0}^9 (J_i + m_i \cdot x_i^2) + \sum_{i=10}^{16} \sum_{j=1}^n m_{ij} \cdot x_{ij}^2 \right) \cdot \frac{d^2}{dt^2} \theta_0(t) \end{aligned} \quad (34)$$

In summary, the structural moment of the wing skeleton is:

$$dM_{total_stru,x} = dM_{w2f-x} + dM_{i2f-x} \quad (35)$$

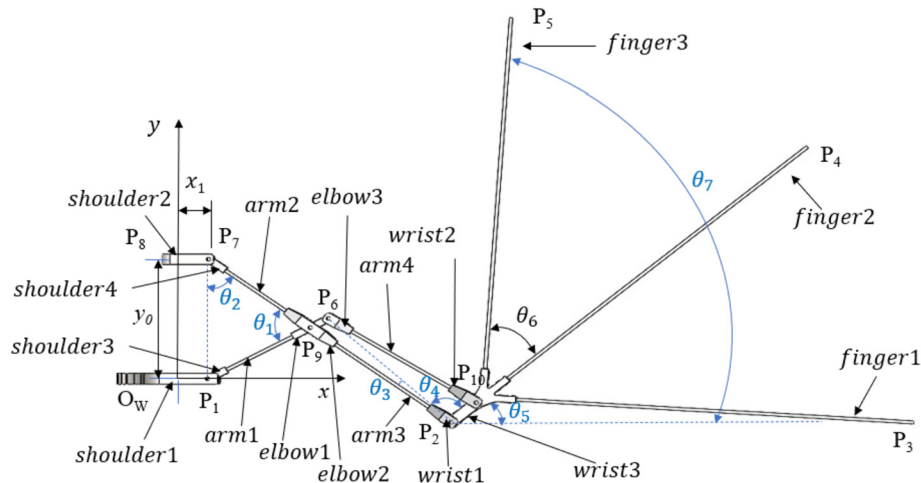


Fig. 10. Structure diagram of each part of wing skeleton.

4. Wing geometry and kinematics optimization

4.1. Power density model

Whether the bat-like robot is designed to meet a certain range and endurance or to carry additional fixed or variable loads, such as avionics sensors and lithium batteries, it is necessary to optimize the geometry and kinematics of the wing to minimize energy consumption.

This paper adopts and extends the power consumption model proposed by Berman and Wang et al. [29]. In addition, the torsional angular velocity around the leading edge axis is too small, almost 0, so the torsional power consumption of the wing around the leading edge axis is ignored. According to the aerodynamic model in Section 3.2 and the structural mechanics model in Section 3.3, the power model of the wing can be derived as follows:

$$dP_{total_aero} = dM_{total_aero} \cdot \omega_{total} = dM_{total_aero,x} \cdot \omega_{\theta} \cdot \cos\varphi \quad (36)$$

$$dP_{total_stru} = dM_{total_stru} \cdot \omega_{total} = dM_{total_stru,x} \cdot \omega_{\theta} \cdot \cos\varphi \quad (37)$$

So, the power density is:

$$P^* = \frac{P_{total_aero} + P_{total_stru}}{Weight} \quad (38)$$

Where, *Weight* is the weight of the bat-like robot.

4.2. Formulation of the optimization problem

The optimization problem is to search for the optimal WGP and WKP of the bat-like robot wings to minimize the power consumption during fixed-altitude flight.

The newly proposed GPSO [30] is used to solve the multi-constrained optimization problem. This algorithm is different from the traditional GPSO algorithm which adopts genetic algorithm (GA) and particle swarm optimization (PSO) in series, it combines PSO and GA through parallel architecture, to make full use of the high efficiency of PSO and the global optimization ability of GA. This algorithm takes PSO as the main body, and GA as a means of updating particles in parallel to complete the optimization task. The parallel relationship refers to that when the algorithm does not fall into the local optimum, PSO is used to update the particles alone. When the algorithm falls into the local optimum, GA is used to select, crossover, and mutation of the particles alone to update the particles, which helps the algorithm jump out of the local optimum. Once the algorithm jumps out of the local optimum, PSO is continued to be used for optimization. Since PSO has memory, GA does not destroy the found optimal value. The parallel structure gives full play to the advantages of PSO efficiency and GA global optimization ability, and avoids the disadvantages of PSO easily falling into local optimum and GA low optimization efficiency. In addition, the GPSO adopts adaptive inertia weight, adaptive mutation parameters, and multi-point crossover operation between particles and personal best value (pbest) to improve the optimization ability of the algorithm. The relevant parameters of GPSO [30] adopted are as follows: particle number 50, crossover rate 0.8, mutation rate 0.05, iteration number 200, maximum inertia weight 0.8, minimum inertia weight 0.4, and learning factor 1.5.

The overall optimization problem can be described as follows:

Parameters to be optimized:

WGP: The length of wing (L_1).

WKP: Amplitude of flapping angle (L_2), flapping frequency (L_3).

$$L_1 = [arm1, arm2, arm3, arm4, finger1, finger2, finger3] \quad (39)$$

$$L_2 = [\theta_0] \quad (40)$$

$$L_3 = [f] \quad (41)$$

$$L = [L_1, L_2, L_3] \quad (42)$$

Objective function:

$$f(x) = \min(P^*(L)) \quad (43)$$

Parameter constraints:

By analyzing the parameter data of 88 bat species from Norberg's statistics [31], the parameter constraints of this paper are obtained:

$$0.02 \leq arm1 = arm2 \leq 0.07 \quad (44)$$

$$0.05 \leq arm3 = arm4 \leq 0.09 \quad (45)$$

$$0.1 \leq finger1 = finger2 \leq 0.2 \quad (46)$$

$$0.1 \leq finger3 \leq 0.2 \quad (47)$$

$$25 \leq \theta_0 \leq 35 \quad (48)$$

$$3 \leq f \leq 10 \quad (49)$$

Conditional constraints:

The constraint of Re: Re is introduced as a nonlinear constraint to ensure the applicability of the quasi-steady aerodynamic model. Referring to the experimental data used by Singh [32], Re in this paper is [0.5e5, 5e5].

$$0.5 \times 10^5 \leq Re = \frac{v_{f\infty} \cdot \left(\sqrt{(arm2 + arm3)^2 - y_0^2} + finger1 \right)}{v_{air}} \leq 5 \times 10^5 \quad (50)$$

The constraint of AR: This paper refers to AR data of 88 bat species in Norberg [31] and applies it to the optimization constraints in this paper. The display ratio of bat single wing in this paper is about 3. The AR used by Singh for calculation is about 2.8 [32]. In consideration of the universality of optimization analysis and feasibility design in engineering, the AR is selected as [2.5, 3.5] in this paper.

$$2.5 \leq AR = \frac{\left(\sqrt{(arm2 + arm3)^2 - y_0^2} + finger1 \right)^2}{0.5 \cdot y_0 \cdot finger1 + y_0 \cdot \sqrt{(arm2 + arm3)^2 - y_0^2}} \leq 3.5 \quad (51)$$

The constraint of St: Taylor [33] predicted that the St of bats are between 0.2 and 0.6. This paper uses the data predicted by Taylor.

$$0.2 \leq St = \frac{f \cdot \left(\sqrt{(arm2 + arm3)^2 - y_0^2} + finger1 \right)}{v_{f\infty}} \leq 0.6 \quad (52)$$

The constraint of LtoW: In order to realize the function of fixed-altitude flight, the bionic bat-like robot involved in this paper needs the LtoW of the model to be 1. However, considering the stability of flight, the LtoW is set as:

$$1 \leq LtoW = \frac{Lift}{Weight} \leq 1 + \delta \quad (53)$$

where δ is the relative error, $\delta = 10^{-8}$ [29].

Table 1

Comparison of geometric parameters and motion parameters.

	Bat parameters	Optimized parameters of the bat-like robot
Humerus (Arm1) length, m	0.038*	0.0508
Radius (Arm3) length, m	0.068*	0.0807
Digits (Finger1) length, m	0.12*	0.1763
Digits (Finger3) length, m	0.12*	0.1656
Flapping amplitude, °	≈35*	27.5
Maximum flapping frequency, Hz	≈10*	10
Wing span of single wing, m	0.3*	0.3404
Mean wing chord, m	0.12*	0.1276
Weight, kg	0.16*	0.2

* Actual WGP and WKP of bats [14].

Table 2

Performance comparison during fixed-altitude flight.

	Original results of bat-like robot	Optimized results of bat-like robot
Flapping frequency at fixed-altitude flight, Hz	7	5
Re	0.89e5	1.1501e5
AR of single wing	2.5	2.6688
St	0.477	0.3404
LtoW-1	0.174	8.1189e-8
Power density, w/kg	50.8831	24.9841

5. Discussion and analysis of results

5.1. Comparative analysis of optimization results

To achieve the goal of minimum energy consumption when the bat-like robot flies at a fixed altitude, a new GPSO is used to optimize the WGP and WKP of the robot by combining the power density model mentioned above with various parameter constraints and conditional constraints. The optimized results are shown in Table 1 and Table 2. Table 1 shows the comparison between the WGP and WKP of the original bat and the optimized bat-like robot. These parameters affect the LtoW and energy density during flight. Table 2 shows the estimated performance parameters of bats and the performance parameters of the optimized bat-like robot at fixed-altitude flight. After optimization, the flapping frequency for maintaining fixed-altitude flight is 5 Hz, which is lower than the original parameter frequency of 7 Hz, and the low frequency will greatly reduce the energy consumption (as explained in the parameter sensitivity analysis); The optimized Re and AR are larger than those of the original parameters; The St after optimization is smaller than the that before optimization. The optimized LtoW is infinitely close to 1, which can better maintain fixed-altitude flight than the original parameter model; The power density of the original parameter is 50.8831 w/kg, and the power density after parameter optimization is 24.9841 w/kg, which is significantly smaller than the power density of the original parameters.

The optimized power components and contribution of each aerodynamic model, as shown in Fig. 11. Fig. 11(a) shows the various parts of the power of change in the stage of flapping motion, including translational circulation power, rotational circulation power, aerodynamic damping power, added-mass power, resultant aerodynamic force power, gravity moment power, inertia moment power, and overall flapping power. Among them, the translational circulation power occupies the dominant position, the aerodynamic damping power is in the secondary dominant position, and the rotational circulation power has the smallest contribution. To more clearly show the influence of each aerodynamic model on aerodynamic power, the contribution percentage of each aerodynamic model to aerodynamic power is extracted, as shown in Fig. 11(b). It can be intuitively seen from the figure that the

contribution of translational circulation accounted for 57.59%, the contribution of aerodynamic damping accounted for 27.63%, the contribution of added-mass force accounted for 13.34%, and the contribution of rotational circulation accounted for 1.44%.

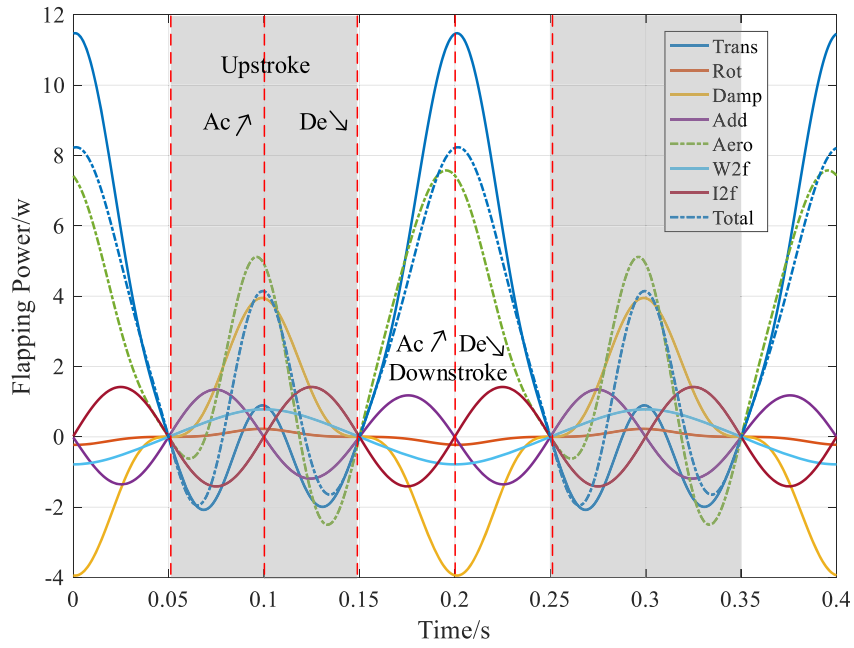
In Fig. 11, the variation of each power is further analyzed in detail, and the curve is divided into two regions. The region with a gray background indicates that the wing is in the upstroke stage, and the region with a white background indicates that the wing is in the downstroke stage. In addition, the upstroke and downstroke stages are divided into the acceleration stage (Ac) and deceleration stage (De). The power curve in a period of 0.05 s ~ 0.25 s will be taken as an example: In the acceleration upstroke stage, the power consumed by the translational circulation is first negative and then positive. This is because, at the initial time of the upstroke, the flapping velocity is 0, so the freestream flow velocity dominates the movement, resulting in negative power consumption. As the flapping velocity gradually increases, the flapping velocity gradually dominates, so the consumed power is gradually positive. The power consumption of rotational circulation, aerodynamic damping, and gravity moment is positive, and increases with the increase of flapping velocity. The power consumed by added-mass is positive and increases first and then decreases with the increase of flapping velocity. The power consumed by inertia moment is negative, and increases first and then decreases with the increase of flapping velocity. Similarly, the change of the power of each part in the deceleration upstroke stage is almost opposite to the change of the power in the acceleration upstroke stage, that is, the power consumed by the translational circulation is positive first and then negative, and the power consumed by the rotational circulation, aerodynamic damping and gravity moment is positive, and decreases with the decrease of the flapping velocity. The power consumed by the added-mass is negative, and increases first and then decreases with the decrease of the flapping velocity. The power consumed by inertia moment is positive, and increases first and then decreases with the decrease of flapping velocity.

In the acceleration downstroke stage, the power consumed by the translational circulation is always positive and increases with the increase of the flapping velocity. The power consumption of rotational circulation, aerodynamic damping, and gravity moment is negative and increases with the increase of flapping velocity. The power consumed by added-mass is positive, and increases first and then decreases with the increase of flapping velocity. The power consumed by inertia moment is negative, and increases first and then decreases with the increase of flapping velocity. Similarly, the change of the power of each part in the deceleration phase is opposite to the change of the power in the acceleration phase, that is, the power consumed by the translational circulation is positive and decreases with the decrease of the flapping velocity. The power consumption of circulation, aerodynamic damping, and gravity moment is negative and decreases with the decrease of flapping velocity. The power consumed by the added-mass is negative, and increases first and then decreases with the decrease of the flapping velocity. The power consumed by inertia moment is positive, and increases first and then decreases with the decrease of flapping velocity.

To understand the relationship between power consumption and wing motion more easily, it is necessary to clearly understand the relationship between membrane deformation and movement. Fig. 12 shows the deformation state of the membrane in the stage of accelerating downstroke, decelerating downstroke, accelerating upstroke, and decelerating upstroke.

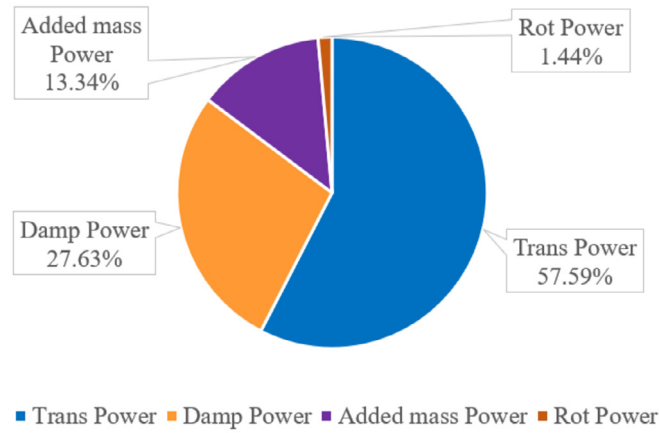
5.2. Sensitivity analysis of parameters

Table 1 presents the optimal WGP and WKP for the bat-like robot to maintain the minimum energy consumption when flying



(a) Flapping power of each component

(The curves in the figure are translational circulation power, rotational circulation power, aerodynamic damping power, added-mass power, resultant aerodynamic force power, gravity moment power, inertia moment power and overall flapping power)



(b) Contribution of each aerodynamic model

Fig. 11. Flapping power and contribution of each aerodynamic model.

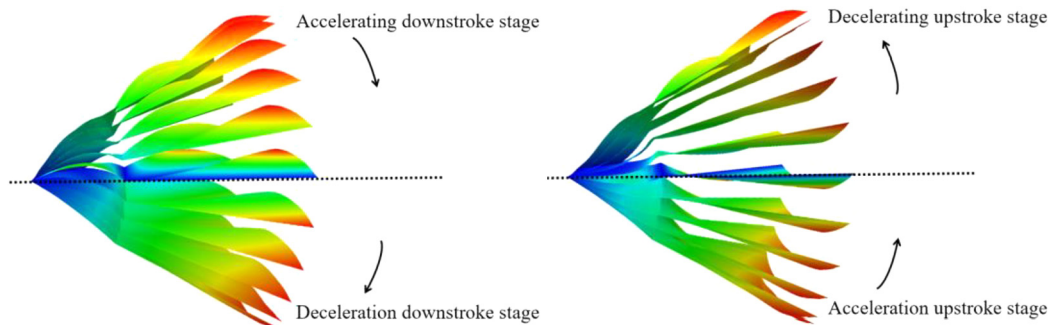


Fig. 12. Relationship between membrane deformation and movement.

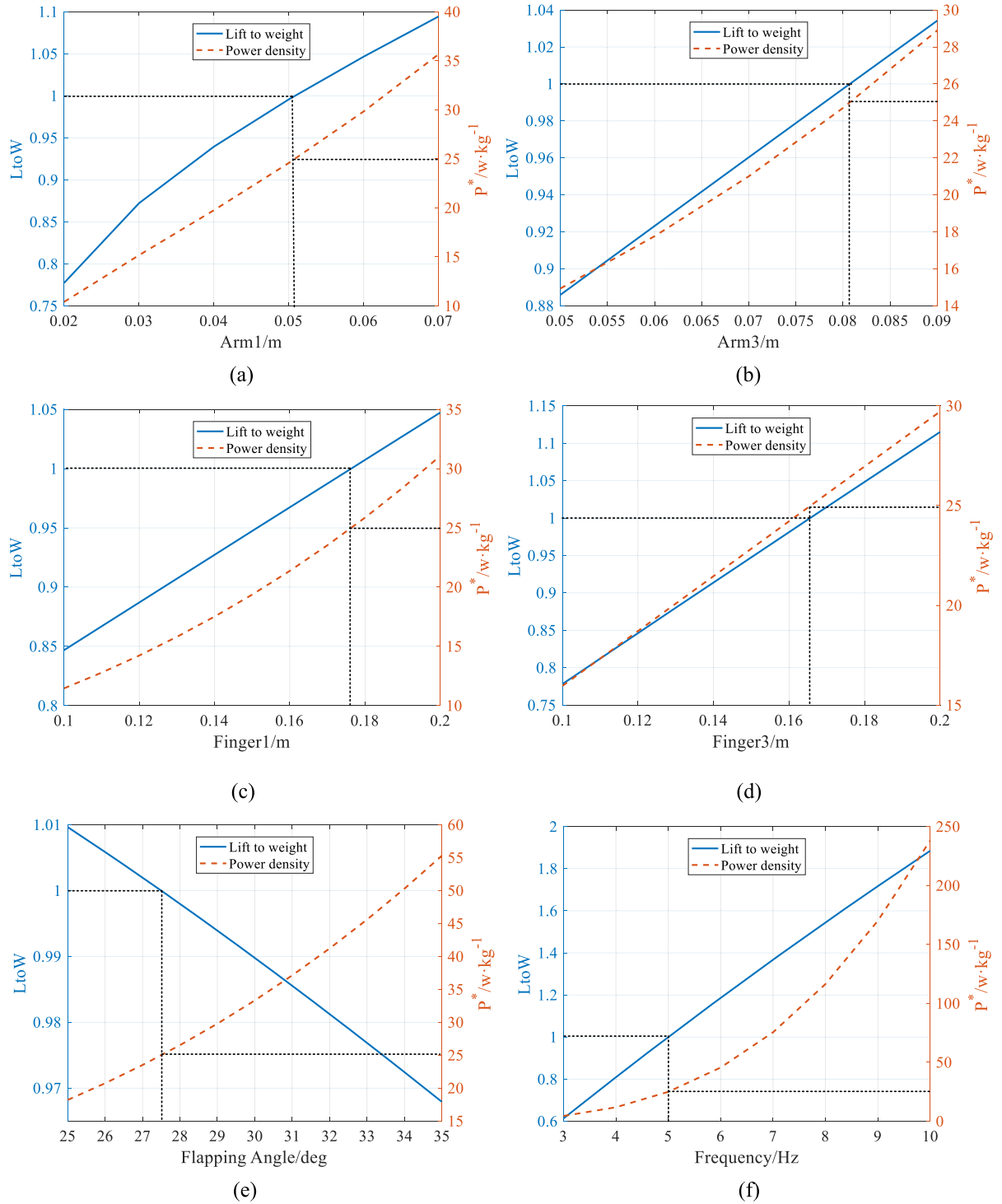


Fig. 13. Sensitivity analysis of parameter sequence constructed by combining optimal WGP and WKP of bat-like robot.

at a fixed-altitude. Next, we need to analyze the influence of various parameters on the LtoW and power density, to explore the influence of WGP and WKP on maintaining the energy consumption of the bat-like robot. To accurately analyze the effect of each parameter, it is necessary to keep the other parameters unchanged (keeping the optimal solution) and carry out a single parameter sensitivity analysis.

Fig. 13 shows the sensitivity analysis diagram of each parameter in the limit range, where the abscissa represents the change of each parameter in the limit range, the blue coordinate axis and curve represent the LtoW, and the red coordinate axis and curve

represent the power density (P^*). Fig. 13 (a) ~ (d) are the sensitivity analysis diagrams of 4 WGP, (e) ~ (f) are the sensitivity analysis diagram of 2 WKP. It can be seen from (a) ~ (d) that the LtoW and power density increase monotonically with the increase of Arm1, Arm3, Finger1, and Finger3. Arm3 and Finger1 affect the aerodynamic forces by changing the wingspan. Finger3 affects the aerodynamic force by changing the chord length. However, Arm1 mainly affects aerodynamic force by affecting the wingspan and the amplitude of wing deformation movement, so the influence of Arm1 on the LtoW is nonlinear, while the influence of Arm3, Finger1, and Finger3 on the LtoW are almost linear. In addition,

the curve change in the diagram also illustrates the contradiction between the LtoW and the power density, that is, increasing the LtoW will inevitably lead to the consumption of additional energy. Therefore, for the optimization of WGP, the WGP with the minimum energy consumption in fixed-altitude flight ($LtoW = 1$) can be found by the optimization algorithm. From (e) ~ (f), it can be seen that the LtoW decreases monotonously with the increase of the amplitude of the flapping angle, increases monotonously with the increase of the flapping frequency, and almost changes linearly. The power density increases nonlinearly with the increase of flapping angle and flapping frequency, especially the flapping frequency factor. When the flapping frequency increases gradually, the nonlinearity of the power density curve increases sharply. By synthesizing all the above influencing factors, WGP and WKP shown in Table 1 are finally obtained to ensure that the bat-like robot maintains the minimum energy consumption when flying at a fixed-altitude.

6. Conclusion

In this paper, to improve the flight endurance of the bat-like robot, GPSO is used to optimize the WGP and WKP of the flapping robot, to meet the requirement of minimum energy consumption during fixed-altitude flight. Before elaborating the optimization problem, firstly, the structural characteristics, motion law, and flexible deformation of bat wings are analyzed in detail, and the chord length and passive torsion angle calculation models of wing membranes during flapping are established. Then, a revised quasi-steady model of aerodynamic forces and moments is developed based on the previous quasi-steady aerodynamic model by additionally introducing passive torsional circulation moment and the aerodynamic damping moment around the leading edge axis, at the same time, the influence of gravity moment and inertia moment of the wing is considered, and the dynamic model of wing flapping is further improved. Furthermore, based on the power density model, the optimization problem of WGP and WKP to minimize the energy consumption of fixed-altitude flight is established with the constraints of LtoW, Re, AR, St, and the boundary of each parameter. Then the GPSO is used to optimize WGP and WKP to minimize the energy consumption of the bat-like robot at a fixed-altitude flight. The optimized aerodynamic force and power density are compared with the estimated value of the actual bat during fixed-altitude flight. The results present that the error of the optimized LtoW can be reduced from 0.174 to $8E-8$, and the energy density can be reduced from 50.8831 W/kg to 24.9841 W/kg, indicating that the optimized parameters can obtain more stable aerodynamic force and reduce the energy consumption by half. Finally, through the analysis of the sensitivity of the parameters, it is concluded that there is a contradiction between the LtoW and the power density for the adjustment of WGP parameters, that is, increasing the LtoW will inevitably lead to the consumption of additional energy, and that the adjustment of WKP parameters will have a strong nonlinear effect on the power density. The research in this paper solves the problem of extra energy consumption during fixed-altitude flight of the bat-like robot, which provides a theoretical basis for the structural optimization design of the bat-like robot, and is conducive to improving the flight performance of the bat-like robot.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgement

This work is supported by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China (No. 51521003), and HIT Wuhu Robot Technology Research Institute (No. HIT-CXY-CMP2-RVJDT-21-01).

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